

Agent Crate Architecture

crates/agent — Der AI-Agent-Layer von Zed

Layer-Stack

ENTRY POINT

agent.rs — Public API & Re-Exports

Root-Modul. Exportiert Thread, Tools, Permissions, Templates, NativeAgentServer. Bindet Context-Server und ACP-Thread-Protokoll ein.

acp_thread

context_server

language_model

CORE RUNTIME

thread.rs — Agent-Thread & Turn-Loop

Zentrale Laufzeit: Message-Historie, Completion-Loop, Tool-Dispatch, Compaction, Cancellation.
>8000 Zeilen — das Herz des Agenten.

tools

tool_permissions

templates

sandboxing

agent_settings

PERSISTENCE

db.rs — SQLite Database

Speichert Threads & Messages persistent via SQLite. Wird von thread_store.rs verwaltet.

sqlite

acp_thread

THREAD MANAGEMENT

thread_store.rs — Thread Store

Verwaltet alle Agent-Threads: Laden, Speichern, Serialisierung. Brücke zwischen Thread-Laufzeit und DB.

db

thread

TOOLS LAYER

tools.rs — Tool Registry & Feature Flags

tools!-Makro registriert 23 Built-in-Tools. Feature-Flag-Gating für LSP, Rename, Subagent-Tools. Template-Generierung für Model-Schema.

[feature_flags](#)[terminal_tool](#)[edit_file_tool](#)[...](#)

SECURITY

tool_permissions.rs — Permission Engine

Hardcoded Security Rules + Regex-basierte Allow/Deny/Confirm-Matrix. Sandbox-Authorization. Symlink-Escape-Detection.

[agent_settings](#)[shell_command_parser](#)

SANDBOX

sandboxing.rs — macOS Seatbelt Integration

OS-Level Sandbox für Terminal-Kommandos. Feature-Flag-gesteuert. Netzwerk/FS-Write-Permission-Escalation.

[feature_flags](#)[sandbox](#)

PROMPTING

templates.rs + templates/ — Handlebars Templates

System-Prompt, Edit-File-Prompt, Diff-Fenced-Format. Handlebars-Rendering mit Projekt-Kontext, Tool-Liste, Skills.

[handlebars](#)[agent_skills](#)

EXTERNAL INTERFACE

native_agent_server.rs — ACP Server

Implementiert das Agent-Client-Protocol (ACP). Verbindet Native UI mit Agent-Backend. Session-Management.

[acp_thread](#)[thread_store](#)

UTILITY

outline.rs + pattern_extraction.rs

File-Outline-Generierung für Read-File-Tool. Pattern-Extraction für Edit-Sessions und Code-Analyse.

Turn Execution Flow

1

Build Request

System-Prompt + Messages + Tools → LanguageModelRequest. Auto-Compact-Prüfung.

2

Stream & Dispatch

LLM-Stream → Parse Tool-Calls → Permission-Check → Run Tools. Parallel für independent scopes.

3

Loop or End

Tool-Results → nächste Iteration oder EndTurn/Refusal/MaxTokens. Retry-Logik.

Tools (23 Built-in)

F **read_file** — Lese Dateien mit Zeilennummern

F **write_file** — Schreibe/Überschreibe Dateien

F **edit_file** — Fuzzy-Edit mit SEARCH/REPLACE

F **delete_path** — Lösche Dateien/Ordner

F **copy_path / move_path** — Kopiere/Verschiebe

F **create_directory** — Erstelle Ordner

S **grep** — Regex-Suche im Code

S **find_path** — Datei-Pfad-Suche

S **list_directory** — Ordnerinhalt auflisten

T **terminal** — Shell-Kommandos ausführen

T **fetch** — HTTP-Requests

A **spawn_agent** — Subagent starten

A **create_thread** — Thread erstellen

A **skill** — Skill laden & aktivieren

S **diagnostics** — Fehler/Warnungen

S **find_references** — Symbol-Referenzen

S **go_to_definition** — Symbol-Definition

A **get_code_actions** — LSP Code Actions

A **apply_code_action** — Code Action anwenden

A **rename** — Symbol umbenennen

S **web_search** — Web-Suche

A **list_agents_and_models** — Agent/Model-Auflistung

Key Dependencies

acp_thread

Agent-Client-Protocol: Session, Events, Tool-Call-Updates, Retry-Status

language_model

LLM-Abstraktion: Request/Response, Completion-Stream, Token-Usage, Tool-Use

agent_settings

Profile, Auto-Compact, Tool-Permissions, Sandbox-Permissions

context_server

Externe Tools via MCP/Agent-Server-Protokoll

project

Worktree-Zugriff, Buffer-Management, File-System

feature_flags

Sandboxing, LSP-Tools, Rename, Create-Thread Feature-Gating

agent_skills

Skill-Discovery: Built-in, Global (~/.agents), Project (.agents)